



PONTIFICIA UNIVERSIDAD CATÓLICA DE CHILE

ESCUELA DE INGENIERÍA

Departamento de Ingeniería Química y Bioprocesos (DIQB)

www.ing.puc.cl/iig

Course Title: Statistical Design, Optimization, and Multivariate Analysis

Course Code: IIQ3402

Credits: 10

Schedule: Tue and Thu Module 3 (Lectures). Wed Module 5 (Tutorial)

Location: Computational Laboratory 1, Raúl Devés Building

Prerequisites: Master's or Doctoral Program in Engineering Sciences

Semester: First Semester, 2025

Instructor: Pedro Saa (PS) – psaa@uc.cl

Teaching Assistant: Gabriel Miranda (GM) – ggmiranda@uc.cl

Course Description

Statistical data analysis is a fundamental component of any research project across scientific disciplines. The appropriate use of statistical tools enables researchers to explore data, test hypotheses, and design experiments to address scientific questions. With today's computational power, it is possible to conduct sophisticated analyses that offer robust answers in a fraction of the time previously required. This course introduces students to exploratory data analysis and experimental planning using Python-based scientific computing tools. Key areas covered include:

- Core statistical concepts guiding the selection of appropriate statistical tests.
- Effective data visualization techniques to enhance communication of findings.
- Application of models to understand relationships among variables.
- Selection of cost-effective experimental designs.
- Development of reproducible data analyses to support replicability and future research.

Learning Objectives

By the end of the course, students will be able to:

- Understand statistical concepts and assumptions underpinning various methods.
- Formulate hypotheses and design experiments for testing them.
- Select and apply univariate and multivariate statistical tests.
- Analyze data and design experiments using Python-based tools.
- Create effective visual representations of statistical results.
- Apply regression models to assess relationships among variables.

- Grasp the four principles of experimental design: control, replicate, block, randomize.
- Select appropriate experimental designs based on scientific problems.
- Use response surface methodology to optimize experiments and processes.
- Implement multivariate analysis tools (PCA, clustering) for data interpretation and design support.

Contents

1. Introduction to Data
 - Data language and types of variables
 - Observational vs. experimental studies
 - Random sampling vs. random assignment
 - Introduction to experimental design principles
2. Exploratory Data Analysis
 - Categorical variables: bar charts
 - Numerical variables: histograms, boxplots, scatterplots, QQ-plots
 - Descriptive statistics and distributional analysis
 - Robust statistics and outlier detection
3. Statistical Inference
 - Confidence intervals
 - Hypothesis testing
 - Mean comparisons: t-tests (independent and paired)
 - Sample size and statistical power
 - ANOVA
 - Categorical data comparison: proportion analysis and χ^2 tests
4. Regression Analysis
 - Correlation
 - Simple and multiple linear regression
5. Experimental Design and Optimization
 - Experimentation strategies
 - Completely randomized designs
 - Factorial randomized designs
 - Randomized block designs
 - Fractional factorial designs
 - Response surface analysis and optimization
6. Multivariate Analysis
 - Principal Component Analysis (PCA)
 - Cluster analysis

Teaching Methodology

- In-person lectures introducing theoretical frameworks.
- Hands-on tutorials applying concepts to case studies using Python.
- Online practice focused on programming, data analysis, and statistics in Python.

Evaluation Criteria

Quizzes (*Average Grade: C*)

Short multiple-choice assessments (12 total) based on lecture content and Python exercises. Administered at the beginning or end of each class. The 10 highest grades will be counted.

Assignments (*Average Grade: T*)

Two analytical tasks (individual or in pairs) involving realistic data problems. Solutions must be reproducible and submitted in Jupyter Notebook format, along with a comprehensive report. Provided at least 5 weeks in advance.

Final Project (*Grade: P*)

An individual project measuring the student's ability to integrate and communicate course concepts. Includes:

- **Proposal:** One-page project outline (20%)
- **Report:** Concise paper-style write-up with theory, methodology, and statistical design (50%)
- **Presentation:** 10-minute overview of research and key findings (30%)

Final Grade

Final Grade (NF) = $0.2C + 0.4T + 0.4P$

A minimum mark of 3.95 is required to approve the course.

Consultation

General course inquiries should be posted in the Canvas discussion forum. For personal matters, contact the instructor or teaching assistant directly.

Code of Honor

The course adheres to the Honor Code of the School of Engineering, which is binding. All evaluated work must be original. If collaboration is permitted, proper attribution is required. Students must be familiar with the online version of the Honor Code: <https://integridadacademica.uc.cl>

Bibliography

Minimal

- Maiti, J. (2022). Multivariate Statistical Modeling in Engineering and Management. CRC Press
- Montgomery, D. (2012). Design and Analysis of Experiments. 8th Ed., Wiley
- Montgomery, D., Runger, G. (2011). Applied Statistics and Probability for Engineers. 5th Ed., Wiley

Recommended Online Python Courses

- [Data Analysis with Python – IBM](#)

- [Statistics for Data Science with Python – IBM](#)
- [Basic Statistics in Python – IBM](#)
- [Data Visualization with Python – IBM](#)

Supplementary

- Dean, A., Voss, D., Draguljic, D. (2017). *Design and Analysis of Experiments*. 2nd Ed., Springer